Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	08 April 2026
Team ID	SWTID-2026-8559
Project Name	Enforcing Mandatory Fields Using UI Policies and Migra ng Changes with Update Sets
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture of this project is designed to enforce mandatory fields dynamically using UI Policies and manage configuration changes using Update Sets in ServiceNow.
It ensures real-time validation, improved user interaction, and safe migration across environments.

Architecture Layers

1. User Interaction Layer

- Users (System Admin / IT Support Agent) interact with ServiceNow forms (e.g., Incident Form)
- Data is entered during record creation and updates

2. UI Policy Layer

- UI Policies are configured to:
  - Enforce mandatory fields
  - Control field visibility
  - Set fields as read-only
- Conditions are applied dynamically based on user inputs

3. Validation Layer

- Real-time validation ensures:
  - Required fields are filled
  - Incorrect or incomplete data is prevented
- Error messages guide users for correct input

4. Update Set Layer

- All configuration changes (UI Policies, form modifications) are captured
- Update Sets act as a **version control system** for tracking changes

5. Migration Layer

- Update Sets are migrated across environments:
  - Development → Testing → Production
- Includes:
  - Conflict detection
  - Dependency validation
  - Controlled deployment

6. Data Storage Layer

- Configuration data and records are stored in ServiceNow tables

- Ensures consistency and integrity across the system

7. Interface Layer

- Enhanced ServiceNow forms provide:
    - Dynamic field behaviour
    - Auto-enforcement of rules
    - Improved user experience

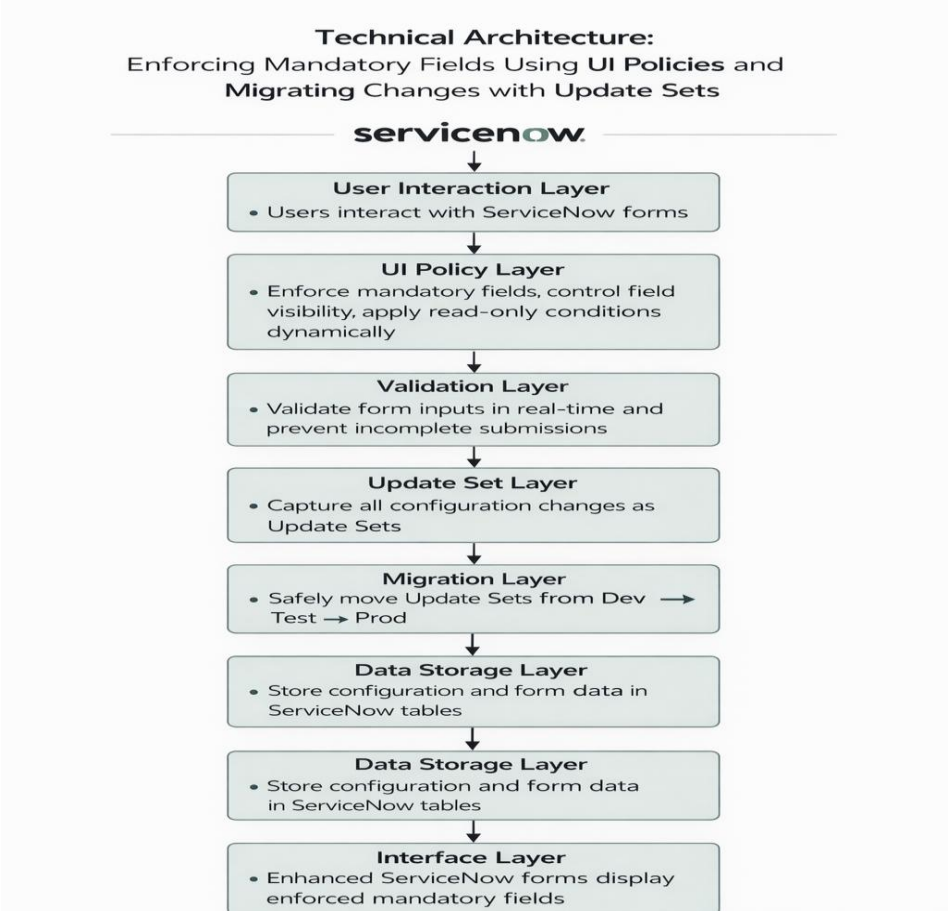
- 

Table 1: Technology Stack Description

S.No	Component	Description	Technology Used
1	User Interface	User interacts with ServiceNow forms where mandatory fields are enforced dynamically	ServiceNow UI (Native Forms & Portal), HTML/CSS (custom UI)

S.No	Component	Description	Technology Used
2	Application Logic-1	UI Policy configuration to enforce mandatory, read-only, and visibility rules	ServiceNow UI Policies (Client-side scripting)
3	Application Logic-2	Real-time validation of form inputs before submission	UI Policies + Client Scripts (JavaScript)
4	Application Logic-3	Capturing and managing configuration changes	Update Sets (ServiceNow Change Management)
5	Application Logic-4	Migration of configurations across environments	Update Set Migration (Dev → Test → Prod)
6	Database	Storage of records and configuration data	ServiceNow Database (MySQL-based)
7	Cloud Database	Cloud-hosted database infrastructure	ServiceNow Cloud Platform (AWS/GCP)
8	File Storage	Storage of attachments and configuration-related data	ServiceNow Attachment Storage
9	External API-1	Integration support for system communication (optional)	ServiceNow REST API
10	Machine Learning	Not required (rule-based UI Policy system)	Not Applicable
11	Infrastructure	Application deployment and hosting	ServiceNow SaaS Cloud Platform

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology Used
1	Open-Source Frameworks	Frameworks used in the project	ServiceNow (Proprietary), JavaScript (for scripting)

S.No	Characteristics	Description	Technology Used
2	Security Implementations	Data protection and access control	ServiceNow ACLs, RBAC, HTTPS/TLS, OAuth 2.0
3	Scalable Architecture	Ability to handle growth and large usage	Multi-tier architecture (UI → Application → Database), auto-scaling cloud
4	Availability	System uptime and reliability	99.99% SLA, multi-data center support, load balancing
5	Performance	System efficiency and response time	Real-time UI Policy execution, minimal server load, optimized form validation